Wherein lies the complexity of interferometric detectors? After all, the basic idea, as described in Figure 10.6, looks reasonably simple.

In fact, Figure 10.6 is a gross oversimplification because it ignores an enormous number of pitfalls. The tricks required to avoid these pitfalls make an interferometer into a very complex instrument. For example, the laser beam must point in precisely the right direction and have precisely the right shape and wavelength to fit into the interferometer perfectly; and its wavelength and intensity must not fluctuate. After the beam is split in half, the two beams must bounce back and forth in the two arms not just once as in Figure 10.6, but many times, so as to increase their sensitivity to the wiggling masses' motions, and after these many bounces, they must meet each other perfectly back at the beam splitter. Each mass must be continually controlled so its mirrors point in precisely the right directions and do not swing as a result of vibrations of the floor, and this must be done without masking the mass's gravitational-wave-induced wiggles. To achieve perfection in all these ways, and in many many more, requires continuously monitoring many different pieces of the interferometer and its light beams, and continuously applying feedback forces to keep them perfect.

One gets some impression of these complications from a photograph (Figure 10.7) of a 40-meter-Iong *prototype* interferometric detector that Drever's team has built at Caltech—a prototype which itself is far simpler than the full-scale, several-kilometer-Iong interferometers that are required for success.